

Investment Screen — DOS App Probe Pack

Per%Sense web port · Plan B reconstruction · 2026-07-22

Why: the Investment screen's compute unit is absent from every source copy in this repo (verified 2026-07-22: no `INVEST*.pas` in `dos_source` / `win_source` / `src_documented` / oracle units; no unit registers `EnterProc[iINV]`; the Delphi .dpr compiles no investment unit). The port therefore reconstructs the engine from the running DOS app. The manual scans already pin the core semantics — chapter 7 p.91: the screen ****equates the value of Funds IN (deposits) with Funds OUT (withdrawals)****, solving for any single blank dollar amount or the interest rate (IRR), optionally in inflation-adjusted constant dollars; and ch.1 p.3: results are **identical to the Present Value screen's** — so the engine is the already-ported PV machinery with a two-sided balance equation. These probes pin down the remaining conventions (rate convention, inflation compounding, basis handling, dispatch/refusals).

BIGGEST-VALUE ITEM — before any probes: scan/photograph ****manual pages 92–105**** (the rest of chapter 7: Financial Planning examples p.93, Investment Analysis examples p.95, Lease or Purchase p.100). Worked examples with printed numbers are golden tests for free and may make several probes unnecessary.

How to run: DOSBox, Alt-I to the Investment screen. ****Restart the app (or File→New) before every probe**** — we have confirmed the DOS app can carry stale session state into solves (discrepancies.md §33). Default Computational Settings (360 basis) unless a probe says otherwise. Dates below are M/D/YYYY; enter years per the app's century setting. For every probe, photograph the full screen after pressing Enter — the picture is the record; no transcription needed.

D — discovery probes (behavior, not numbers)

- **D1** Empty screen, press Enter. Then only a rate (6), Enter. Record every message. (What does it consider under-determined?)
- **D2** One deposit row: Date 1/1/2020, Amount 10000 — cursor into the dollars cell: what does it display, and which key cycles it (space?)? Record the exact captions of its states (we expect Current/Constant forms).
- **D3** Press Alt-F1 (Examples) on this screen. Photograph the example LIST, then run each example and photograph its result screen. (These are built-in worked cases — more free goldens.)
- **D4** Enter the Q1 worksheet but leave TWO amounts blank → message? Then no blanks at all (all cells filled consistently, then inconsistently — e.g. withdrawal 99,999) → does it refuse, recompute something, or accept?
- **D5** Photograph the Settings line at the bottom of this screen after

changing Basis to 365 and back (which Computational Settings does this screen honor?). Also: does Ctrl-T produce a table on this screen? Photograph it.

Q — quantitative probes (each discriminates hypotheses)

Rate 6, Inflation blank, Constant date blank, 360 basis unless noted.

H1 = rate cell holds a TRUE (continuously-compounded) rate; H2 = a LOAN rate (monthly compounding). The recorded value tells us which (or neither).

#	Worksheet	Blank (solve)	H1 predicts	H2 predicts
Q1	Deposit lump 10,000 @ 1/1/2020	Withdrawal lump amount @ 1/1/2030	18,221.19	18,193.97
Q2	Withdrawal lump 20,000 @ 1/1/2030	Deposit lump amount @ 1/1/2020	10,976.23	10,992.65
Q3	Deposit 10,000 @ 1/1/2020; Withdrawal 20,000 @ 1/1/2030	Interest rate (IRR)	6.9315	6.9515
Q4	Periodic deposit 100/mo From 1/1/2020 Through 12/1/2029 (PrYr 12)	Withdrawal lump @ 1/1/2030	16,483.52	16,469.87
Q5	Deposit lump 200,000 @ 1/1/2030; Periodic withdrawal From 2/1/2030 Through 1/1/2050 PrYr 12, amount blank	Periodic withdrawal amount	1,434.60	1,432.86
Q6	As Q5 but withdrawal amount 1,000 with Dollars = Constant , Inflation 3, Constant date 1/1/2030, Through 1/1/2040; deposit lump @ 1/1/2030 blank	Deposit amount	103,324.57 (yield-conv. infl) / 103,543.17 (raw-cont. infl)	—
Q7	Q6 with Constant date moved to 1/1/2035 (else identical)	Deposit amount	records whether the anchor shifts amounts (predict: $\times \exp(-\text{infl} \cdot 5)$ scaling $\approx 89,036$ under yield-conv.)	—
Q8	Q1 with Basis 365 in Settings	Withdrawal amount	18,220.07 (365.25 day-count on clean dates $\approx$ H1)	—
Q9	Q1 but withdrawal date 1/1/2015 (BEFORE the deposit)	Withdrawal amount	7,408.18 (H1 discount back 5y)	7,413.72
Q10	Q1 exactly, but run WITHOUT restarting after Q9	Withdrawal amount	should equal Q1 — if not, the stale-state bug affects this screen too	

Also record for Q1: the exact decimals displayed (2? 4?), and whether the solved cell renders highlighted/soft (computed) like other screens.

What happens next

Manual pages + probe photos come back → the Go engine is implemented over the existing PV primitives, fit to reproduce every probe and manual example to the cent, each becoming a pinned golden test (internal/finance/investment/). Conventions the probes leave ambiguous ship as documented assumptions in docs/discrepancies.md until contradicted.